Group members:

1. MUHAMMAD AFIQ DANIAL BIN ROZAIDIE A23CS0117
2. MOHAMED ALIF FATHI BIN ABDUL LATIF A23CS0112

SECJ2013 - DSA

2024/2025-1

QUIZ 1

Given that two functions (**funA** and **funB**) have been defined as follows:

```
void funA(int n) {
  cout << "funA - " << n << "\n";
  n = n + 1;
  if (n < 5) {
    if (n % 2 == 0) {
      funB(n);
    } else {
      funA(n);
    }
  }
}
```

```
void funB(int n) {
  cout << "funB - " << n << "\n";
  n = n + 1;
  if (n < 5) {
    if (n % 2 == 0) {
      funB(n);
    } else {
      funA(n);
    }
  }
}
```

Study the funA and funB functions above. Identify the followings:

1. Instruction to check if the terminal case has been reached: _if (n < 5)__________________
2. Instruction to change the size of the problem so the terminal case can be reached: n = n+1_____________

Trace the recursive calls of the funA and funB functions inside the following main function of a C++ program:

```cpp
int main() {
  funA(1);
  return 0;
}
```

```cpp
void funA(int n) {
  cout << "funA - " << n << "\n";
  n = n + 1;
  if (n < 5) {
    if (n % 2 == 0) {
      funB(n);
    } else {
      funA(n);
    }
  }
}
```

```cpp
void funB(int n) {
  cout << "funB - " << n << "\n";
  n = n + 1;
  if (n < 5) {
    if (n % 2 == 0) {
      funB(n);
    } else {
      funA(n);
    }
  }
}
```

funA(1)

```
void funA(int 1) {
cout << "funA - " << 1 << "\n";
n = 1 + 1;
if (2 < 5) {
 if (2 % 2 == 0){
funB(2);
}
}
}
```

Output:
funA - 1

```
void funB(int 2) {
cout << "funB - " << 2 << "\n";
n = 2 + 1;
if (3 < 5) {
funA(3)
}
}
```

Output:
funA - 1
funB - 2

```
void funA(int 3) {
cout << "funA - " << 3 << "\n";
n = 3 + 1;
if (4 < 5) {
 if (2 % 2 == 0) {
funB(4);
}
}
}
```

Output:
funA - 1
funB - 2
funA - 3

```
void funB(int 4) {
cout << "funB - " << 4 << "\n";
n = 4 + 1; }
```

Output:
funA - 1
funB - 2
funA - 3
funB - 4